

Laser-guided missiles

Laser-guided missiles use both beam riding as well as semi-active laser-guidance concepts. RBS-70/ RBS-70NG and LAHAT are examples of laser beam-riding missiles. These were briefly described in part 1 of the article. AGM-114 Hellfire-II is a combat-proven tactical surface-to-surface and air-to-surface missile system that uses semi-active laser homing. Fig. 14 shows Hellfire-II fired from a land vehicle.

Hellfire family comprises Longbow Hellfire and Hellfire-II missiles. Hellfire-II missile has a maximum range of 7km (direct fire) and 8km (indirect fire). The missile can be

launched from multiple air, sea and ground platforms, either in autonomous mode or with remote designation. A variant designated AGM-114L uses millimetre-wave radar guidance. Manufactured by Lockheed Martin and introduced into service in 1984, its primary use is as air-to-surface to engage and defeat individual static or moving advanced armour, mechanised or vehicular targets, patrol craft, buildings and bunkers. AGM-114K, AGM-114M, AGM-114N and AGM-114R are laser-guided variants.

New developments

While laser-guided bomb kits continue to improve in terms of hit accuracy, operational range, guidance technology and so on, in recent years, there has been emphasis to improve guidance technology to improve the weapon's performance in adverse weather conditions. This has been made possible by combining laser guidance with global positioning system (GPS)/inertial navigation system (INS). Laser joint direct attack munition (LJDAM) is an example.

Another major development has been the use of guidance technology in smaller ammunition. In the recent past, field trials have shown encouraging results in laser-guided bullets.

Joint direct attack munition (JDAM) is a low-cost guidance kit used to convert existing unguided free-fall bombs into near-precision-guided weapons. The JDAM kit consists of a tail section that contains a GPS/INS and body strakes for

additional stability and lift. JDAM is produced by Boeing.

LJDAM expands the capabilities of the JDAM by combining a laser sensor kit with a JDAM kit. LJDAM has the accuracy of a laser-guided weapon and all-weather capability and longer range of GPS/INS guided weapons. It can precisely hit both stationary and mobile targets. LJDAM has been integrated with GBU-38 and is operational on the US Air Force F-15E and F-16 and the US Navy F/A-18 and A/V-8B platforms. It is planned to integrate LJDAM with GBU-31 and GBU-32. Fig. 15 shows Boeing LJDAM on F-16 fighter aircraft (lowermost weapon in the figure).

Laser-guided bullet development at Sandia National Laboratories is making headlines as it is expected to significantly increase the range of sharp shooting. Modern bullets gain their accuracy from a technique known as rifling, in which the rifle barrel has a series of spiralling grooves etched into it. The spiralled grooves give a spin to the bullet, thereby stabilising its flight path. The laser-guided bullet developed at Sandia National Laboratories is fired from a smooth bore barrel and is stabilised by four steerable fins at its rear (Fig. 16). The fin movement is controlled by a computer chip, which, in turn, is driven by a signal from an optical sensor on the bullet's nose. The intended target is illuminated by a laser beam and the bullet uses steerable fins to adjust its mid-flight trajectory. The operation is similar to that of a laser-guided munition, which makes a laser-guided bullet nothing but a miniature laser-guided munition.

According to one computer simulation, an unguided bullet fired at a target at 800m would miss the target by about 9m. The laser-guided bullet, on the other hand, would cut that inaccuracy to just 20cm. Knowing peculiarities of ballistics, the accuracy gets better for longer ranges.

To be continued...



Fig. 14: Hellfire-II fired from a land vehicle



Fig. 15: LJDAM integrated with F-16C aircraft



Fig. 16: Prototype of the laser-guided bullet developed at Sandia National Lab